Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim:** To recommend an item to the user using content-based filtering approach	
Experiment No: 11	**Date:**	**Enrolment No:92301733054**

Aim: To recommend an item to the user using content-based filtering approach

IDE: Google Colab

Theory:

There are a lot of applications where websites collect data from their users and use that data to predict the likes and dislikes of their users. This allows them to recommend the content that they like. Recommender systems are a way of suggesting or similar items and ideas to a user's specific way of thinking.

Recommender System is different types:

- **Collaborative Filtering:** Collaborative Filtering recommends items based on similarity measures between users and/or items. The basic assumption behind the algorithm is that users with similar interests have common preferences.
- **Content-Based Recommendation:** It is supervised machine learning used to induce a classifier to discriminate between interesting and uninteresting items for the user.

Content-Based Recommendation System: Content-Based systems recommends items to the customer similar to previously high-rated items by the customer. It uses the features and properties of the item. From these properties, it can calculate the similarity between the items.
In a content-based recommendation system, first, we need to create a profile for each item, which represents the properties of those items. From the user profiles are inferred for a particular user. We use these user profiles to recommend the items to the users from the catalog.

Item profile:
In a content-based recommendation system, we need to build a profile for each item, which contains the important properties of each item. For Example, If the movie is an item, then its actors, director, release year, and genre are its important properties, and for the document, the important property is the type of content and set of important words in it.
Let's have a look at how to create an item profile. First, we need to perform the TF-IDF vectorizer, here TF (term frequency) of a word is the number of times it appears in a document and The IDF (inverse document frequency) of a word is the measure of how significant that term is in the whole corpus.

User profile:
The user profile is a vector that describes the user preference. During the creation of the user's profile, we use a utility matrix that describes the relationship between user and item. From this information, the best estimate we can decide which item the user likes, is some aggregation of the profiles of those items.

Advantages and Disadvantages:

- **Advantages:**
    - No need for data on other users when applying to similar users.
    - Able to recommend to users with unique tastes.
    - Able to recommend new & popular items

- Explanations for recommended items.

- **Disadvantages**:
  - Finding the appropriate feature is hard.
  - Doesn't recommend items outside the user profile.

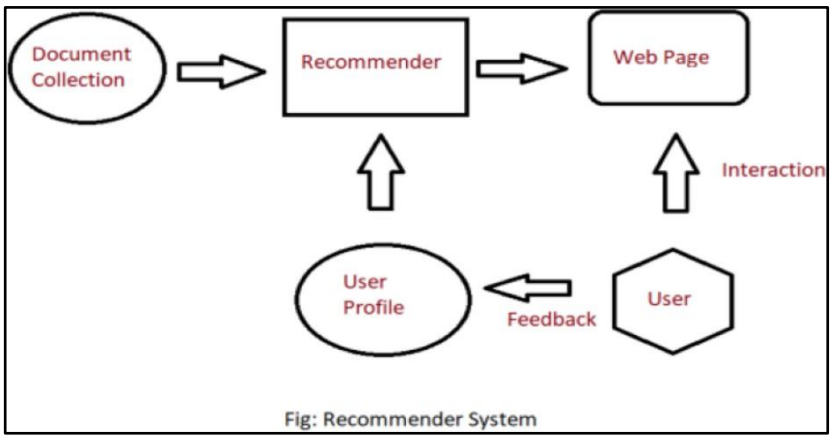

Fig: Recommender System

Pre Lab Exercise:

1. When can you use content-based recommendation system approach?
   - When user preference/history is available
   - When items have well-defined features
   - When you want to recommend similar items based on past choices
   - When other users' data is not available or not needed
   - When personalization is required for individual users
2. Write advantages of content-based filtering approach
   - No need for other users' data
   - Provides personalized recommendations
   - Can recommend new or unseen items
   - Easy to explain recommendations
   - Works well for users with unique tastes
3. Write disadvantages of content-based filtering approach

   - Limited diversity
   - Cold-start problem for new users
   - Requires feature extraction
   - Cannot easily capture changing user interests
   - May lead to over-specialization

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Program (Code):

```
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
df=pd.read_csv("/content/movies.csv")
#preprocessing step
df["genres"]=df["genres"].fillna("")
df["overview"]=df["overview"].fillna("")
df["content"]=df["genres"]+" "+df['overview']
#TF-IDF Vectorizer
tfidf=TfidfVectorizer(stop_words="english")
tfidf_matrix=tfidf.fit_transform(df["content"])
cosine_sim=cosine_similarity(tfidf_matrix,tfidf_matrix)
#droping the duplicates
indices=pd.Series(df.index,index=df["original_title"]).drop_duplicates()
def recommendation_movie(title,topn=5):
  if title not in indices:
    return "Movie is not found!"
  idx=indices[title]
  sim_scores=list(enumerate(cosine_sim[idx]))
  sim_scores=sorted(sim_scores,key=lambda x:x[1],reverse=True)
  sim_scores=sim_scores[1:topn+1]
  movie_indices=[i[0] for i in sim_scores]
  return df["original_title"].iloc[movie_indices]
recommendation_movie("Inception")
```

Results:

	original_title
2897	Cypher
4401	The Helix... Loaded
134	Mission: Impossible - Rogue Nation
1786	Flatliners
2389	Renaissance

dtype: object

Observation: We learned this experiment to recommend items using a content-based filtering approach based on item features like genres and overview. The system uses TF-IDF and cosine similarity to find similar items and provide personalized recommendations. The results show that recommendations are relevant but mostly similar in nature.

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Post Lab Exercise:

1. What changes you feel can be done, to improve the system that you have designed, as a part of in-house lab exercise.

   To improve the content-based recommendation system, more features such as cast, director, and ratings can be included to enhance accuracy. Advanced NLP techniques like Word2Vec or BERT can be used instead of TF-IDF for better understanding of text. A hybrid approach combining collaborative filtering can improve diversity in recommendations. Additionally, incorporating user profiles and feedback can make the system more personalized and effective.

2. Recommend top-10 "series" using IMDB movie dataset (Link: https://www.kaggle.com/datasets/harshitshankhdhar/imdb-dataset-of-top-1000- movies-and-tv-shows). Attach the code with output. Also, write your observation for the same.

Program Code:

```
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

df = pd.read_csv("/content/imdb_top_1000.csv")

df["Genre"] = df["Genre"].fillna("")
df["Overview"] = df["Overview"].fillna("")
df["Director"] = df["Director"].fillna("")
df["Star1"] = df["Star1"].fillna("")
df["Star2"] = df["Star2"].fillna("")
df["Star3"] = df["Star3"].fillna("")
df["Star4"] = df["Star4"].fillna("")

df["content"] = (df["Genre"] + " " +df["Overview"] + " " +df["Director"] + " " df["Star1"] + " " +df["Star2"] + " "
    +df["Star3"] + " " +df["Star4")

tfidf = TfidfVectorizer(stop_words="english")
tfidf_matrix = tfidf.fit_transform(df["content"])

cosine_sim = cosine_similarity(tfidf_matrix, tfidf_matrix)
indices = pd.Series(df.index, index=df["Series_Title"]).drop_duplicates()
def recommend_movie(title, topn=5):
```

```
    if title not in indices:
        return "Movie not found!"

    idx = indices[title]

    sim_scores = list(enumerate(cosine_sim[idx]))
    sim_scores = sorted(sim_scores, key=lambda x: x[1], reverse=True)

    sim_scores = sim_scores[1:topn+1]
    movie_indices = [i[0] for i in sim_scores]

    return df[["Series_Title", "Genre", "IMDB_Rating"]].iloc[movie_indices]
print(recommend_movie("Inception", 5))
```

Output:

```
                Series_Title                     Genre  IMDB_Rating
155          Batman Begins         Action, Adventure          8.2
754  (500) Days of Summer    Comedy, Drama, Romance          7.7
21            Interstellar  Adventure, Drama, Sci-Fi          8.6
934       Mysterious Skin                     Drama          7.6
907                 50/50    Comedy, Drama, Romance          7.6
```